For the use of only a Registered Medical Practitioner or a Hospital or a Laboratory

Gemita RTU 38 mg/ml Injection
Concentrate for Solution for Infusion

DESCRIPTION:

Gemcitabine HCl is a nucleoside metabolic inhibitor that exhibits antitumor activity. Gemcitabine HCl is a white to off-white solid. It is soluble in water, slightly soluble in methanol, and practically insoluble in acetone. Gemcitabine Injection is clear, colourless to light straw colored solution. After dilution, a clear, colorless solution practically free from visible particles.

COMPOSITION:

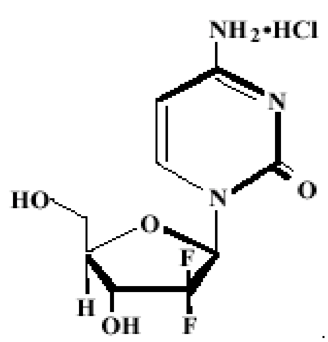
Gemcitabine Injection 38mg/ml

Each mL contains:
Gemcitabine Hydrochloride USP equivalent to
Gemcitabine base ……………………...38 mg
Propylene Glycol USP....……..………..30%v/v
Polyethylene Glycol 400 USNF……….20%v/v
Hydrochloric acid USNF.………………q.s to adjust pH
Sodium hydroxide USNF.......................q.s to adjust pH
Water for Injection USP………………..q.s

CHEMICAL STRUCTURE:

Gemcitabine is a nucleoside metabolic inhibitor that exhibits antitumor activity. Gemcitabine HCl is 2' -deoxy-2',2' -difluorocytidine monohydrochloride (ß-isomer).

The structural formula is as follows:



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PHARMACOLOGY:

Mechanism of action

Gemcitabine kills cells undergoing DNA synthesis and blocks the progression of cells through the G1/S-phase boundary. Gemcitabine is metabolized by nucleoside kinases to diphosphate (dFdCDP) and triphosphate (dFdCTP) nucleosides. Gemcitabine diphosphate inhibits ribonucleotide reductase, an enzyme responsible for catalyzing the reactions that generate deoxynucleoside triphosphates for DNA synthesis, resulting in reductions in deoxynucleotide concentrations, including dCTP. Gemcitabine triphosphate competes with dCTP for incorporation into DNA. The reduction in the intracellular concentration of dCTP by the action of the diphosphate enhances the incorporation of gemcitabine triphosphate into DNA (self-potentiation). After the gemcitabine nucleotide is incorporated into DNA, only one additional nucleotide is added to the growing DNA strands, which eventually results in the initiation of apoptotic cell death.

Pharmacokinetics

<u>Absorption and Distribution</u>

Volume of distribution of gemcitabine was 50 L/m^2 following infusions lasting <70 minutes. For long infusions, the volume of distribution rose to 370 L/m^2.

Gemcitabine pharmacokinetics are linear and are described by a 2-compartment model. Population pharmacokinetic analyses of combined single and multiple dose studies showed that the volume of distribution of gemcitabine was significantly influenced by duration of infusion and gender. Gemcitabine plasma protein binding is negligible.

<u>Metabolism</u>

The active metabolite, gemcitabine triphosphate, can be extracted from peripheral blood mononuclear cells. The half-life of the terminal phase for gemcitabine triphosphate from mononuclear cells ranges from 1.7 to 19.4 hours.

<u>Excretion</u>

Clearance of gemcitabine was affected by age and gender. The lower clearance in women and the elderly results in higher concentrations of gemcitabine for any given dose. Differences in either clearance or volume of distribution based on patient characteristics or the duration of infusion result in changes in half-life and plasma concentrations. Table 1 shows plasma clearance and half-life of gemcitabine following short infusions for typical patients by age and gender. Gemcitabine half-life for short infusions ranged from 42 to 94 minutes, and the value for long infusions varied from 245 to 638 minutes, depending on age and gender, reflecting a greatly increased volume of distribution with longer infusions.

Table 1: Gemcitabine Clearance and Half-Life for the "Typical" Patient

Age	Clearance Men (L/hr/m²)	Clearance Women (L/hr/m²)	Half-life Men (min)	Half-life Women (min)
29	92.2	69.4	42	49
45	75.7	57.0	48	57
65	55.1	41.5	61	73
79	40.7	30.7	79	94

⁺ Half-life for patients receiving a short infusion (<70 min)

INDICATIONS:

Gemcitabine is indicated:
- for treatment of patients with locally advanced or metastatic non-small cell lung cancer (NSCLC).
- for treatment of patients with locally advanced or metastatic adenocarcinoma of the pancreas.
- for treatment of patients with Flurouracil refractory pancreatic cancer.
- alone or in combination with cisplatin, is indicated for treatment of patients with bladder cancer at the invasive stage.
- in combination with paclitaxel, for the treatment of patients with unresectable, locally recurrent or metastatic breast cancer who have - relapsed following adjuvant/neoadjuvant chemotherapy. Prior chemotherapy should have included an anthracycline unless clinically contraindicated.
- in combination with carboplatin, for the treatment of patients with recurrent epithelial ovarian carcinoma, who have relapsed > 6 months following platinum-based therapy'

CONTRAINDICATIONS:

Gemcitabine is contraindicated in those patients with a known hypersensitivity to the gemcitabine.

ADVERSE EFFECTS:

Haematological: Because gemcitabine is a bone marrow suppressant, anaemia, leucopenia and thrombocytopenia can occur as a result of administration of gemcitabine. Febrile neutropenia is also commonly reported.

Gastro-intestinal: Abnormalities of liver transaminase enzymes occur in about two-thirds of patients, but they are usually mild, non-progressive and rarely necessitate stopping treatment. However, gemcitabine should be used with caution in patients with impaired liver function. Nausea and nausea accompanied by vomiting. Diarrhea, constipation, stomatitis and oral toxicity (mainly soreness and erythema) are also commonly reported.

Genito-urinary System: Mild proteinuria and haematuria are reported in approximately half the patients, but are rarely clinically significant, and are not usually associated with any change in serum creatinine or blood urea nitrogen. Gemcitabine should be discontinued at the first signs of any evidence of microangiopathic haemolytic anaemia such as rapidly falling haemoglobin with concomitant thrombocytopenia, elevation of serum bilirubin, serum creatinine, blood urea nitrogen, or LDH. Renal failure may not be reversible even with discontinuation of therapy, and dialysis may be required.

Skin and Appendages: A rash is very commonly seen and is frequently associated with pruritus. The rash is usually mild, not dose-limiting, and responds to local therapy.Severe skin reactions, including desquamation and bullous skin eruptions, have been reported very rarely. Alopecia

Vascular: Clinical sign of peripheral vasculitis and gangrene.

Respiratory: Dyspnoea usually mild, short-lived, rarely dose-limiting, and usually abates without any specific therapy. Bronchospasm after gemcitabine infusion is usually mild and transient, but parenteral therapy may be required. Gemcitabine should not be administered to patients with a known hypersensitivity to this drug. Pulmonary effects, sometimes severe (such as pulmonary edema, interstitial pneumonitis, or adult respiratory distress syndrome (ARDS) The etiology of these effects is unknown. If such effects develop, consideration should be made to discontinuing gemcitabine. Early use of supportive care measures may help ameliorate the condition.

Body as a Whole: Flu-like symptoms are very common. These are usually mild, short-lived, and rarely dose-limiting. Fever, headaches, back pain, chills, myalgia, asthenia, and anorexia are the most commonly reported symptoms. Cough, rhinitis, malaise, sweating and insomnia are also commonly reported. Fever and asthenia are also reported frequently as isolated symptoms. The mechanism of this toxicity is unknown. Reports received indicate that paracetamol may produce symptomatic relief.

Oedema/peripheral oedema is reported by approximately 30% of patients. Some cases of facial oedema have also been reported. Oedema/peripheral oedema is usually mild to moderate, rarely dose-limiting, is sometimes reported as painful and is usually reversible after stopping gemcitabine treatment. The mechanism of this toxicity is unknown. It is not associated with any evidence of cardiac, hepatic or renal failure.

Allergic: Anaphylactoid reaction

Injury, Poisoning, and Procedural Complications: Radiation toxicity and radiation recall reactions.

Cardiovascular: A few cases of hypotension have been reported. Cases of myocardial infarction, congestive heart failure and arrhythmias (predominantly supraventricular in nature) have been reported, but there is no clear evidence that gemcitabine causes cardiac toxicity.

Hepatobiliary: Abnormalities of liver transaminase enzymes (AST and ALT) and alkaline phosphatase occur in about two-thirds of patients, but they are usually mild, non-progressive and rarely necessitate stopping treatment. However, gemcitabine should be used with caution in patients with impaired liver function. Elevations in gamma-glutamyl transferase (GGT) and bilirubin levels have been reported rarely.

Other effects: Somnolence is commonly reported.

Interactions with Other Medicaments:

Radiotherapy: Concurrent (given together or equal to or 7 days apart) - Toxicity associated with this multimodality therapy is dependent on many different factors, including dose of gemcitabine, frequency of gemcitabine administration, dose of radiation, radiotherapy planning technique, the target tissue and target volume. The optimum regimen for safe administration of gemcitabine with therapeutic doses of radiation has not yet been determined.

Radiation injury has been reported on targeted tissues (e.g. oesophagitis, colitis and pneumonitis) in association with both concurrent and non-concurrent use of gemcitabine.

When given in combination with paclitaxel, cisplatin, or carboplatin, the pharmacokinetics of gemcitabine were not altered. Gemcitabine had no effect on paclitaxel pharmacokinetics.

Laboratory Tests: Therapy should be started cautiously in patients with compromised bone marrow function. As with other oncolytics, the possibility of cumulative bone marrow suppression when using combination or sequential chemotherapy should be considered.

Patients receiving gemcitabine should be monitored prior to each dose for platelet, leucocyte and granulocyte counts. Suspension or modification of therapy should be considered when drug-induced marrow depression is detected. For guidelines regarding dose modifications (see Dosage and Administration). Peripheral blood counts may continue to fall after the medicine is stopped.

Laboratory evaluation of renal and hepatic function should be performed periodically. Raised liver transaminases [aspartate aminotransferase (AST) and / alanine aminotransferase (ALT)] and alkaline phosphatase are seen in approximately 60% of the patients. These increases are usually mild, transient and not progressive, and seldom lead to cessation of treatment (see Adverse Effects). Increased bilirubin (WHO toxicity degrees 3 and 4) was observed in 2.6% of the patients. Gemcitabine should be given with caution to patients with impaired hepatic function.

Administration of gemcitabine in patients with concurrent liver metastases or a pre-existing medical history of hepatitis, alcoholism or liver cirrhosis may lead to exacerbation of the underlying hepatic insufficiency.

A few cases of renal failure, including haemolytic uraemic syndrome have been reported (see Adverse Effects). Gemcitabine should be administered with caution to patients with impaired renal function. Gemcitabine treatment should be withdrawn if there is any sign of micro-angiopathic haemolytic anaemia, such as rapidly falling haemoglobin levels with simultaneous thrombocytopenia, elevation of serum bilirubin, serum creatinine, urea or LDH. Renal failure may be irreversible despite withdrawal of the Gemcitabine treatment and may require dialysis.

No incompatibilities have been identified, however, it is not recommended to mix gemcitabine with other medicines.

PRECAUTIONS AND WARNINGS:

Prolongation of the infusion time and the increased dosing frequency have been shown to increase toxicity. In common with other cytotoxic agents, gemcitabine has demonstrated the ability to suppress the bone marrow. Leucopenia, thrombocytopenia and anaemia are expected adverse events. However, myelosuppression is short lived.

Gemcitabine has been reported to cause somnolence. Patients should be cautioned against driving or operating machinery until it is established that they do not become somnolent.

Product is for single use in one patient only.

Patients receiving therapy with gemcitabine must be monitored closely. Laboratory facilities should be available to monitor drug tolerance. Resources to protect and maintain a patient compromised by drug toxicity may be required.

Interstitial pneumonitis together with pulmonary infiltrates has been seen in less than 1% of the patients. In such cases, Gemcitabine Injection treatment must be stopped. Steroids may relieve the symptoms in such situations. Severe rarely fatal pulmonary effects, such as pulmonary oedema, interstitial pneumonitis and acute respiratory distress syndrome (ARDS) have been reported as less common or rare. In such cases, cessation of Gemcitabine Injection treatment is necessary. Starting supportive treatment at an early stage may improve the situation.

Use in pregnancy (Category D†)

Cytotoxic agents can produce spontaneous abortion, foetal loss and birth defects. Gemcitabine Injection must not be used during pregnancy. Women of childbearing age receiving gemcitabine should be advised to avoid becoming pregnant, and to inform the treating physician immediately should this occur.

†Category D: Drugs which have caused, are suspected to have caused or may be expected to cause, an increased incidence of human foetal malformation or irreversible damage. These drugs may also have adverse pharmacological effects.

Use in lactation

It is not known whether the medicine is excreted in human milk. The use of gemcitabine should be avoided in nursing women because of the potential hazard to the infant.

Paediatric Use:

Gemcitabine has been studied in limited Phase 1 and 2 trials in children in a variety of tumour types. These studies did not provide sufficient data to establish the efficacy and safety of gemcitabine in children.

Use in the elderly:

Gemcitabine has been well tolerated in patients over the age of 65. There is no evidence to suggest that dose adjustments are necessary in the elderly, although gemcitabine clearance and half-life are affected by age.

DOSAGE AND ADMINISTRATION:

Non-small cell lung cancer:

Single-agent use:

Adults: The optimum dose schedule for gemcitabine has not been determined. The recommended dose of gemcitabine is 1,000 mg/m^2, given by 30 minute intravenous infusion. This should be repeated once weekly for three weeks, followed by a one week rest period. This four week cycle is then repeated. Dosage reduction with each cycle or dose omission within a cycle may be applied based upon the amount of toxicity experienced by the patient.

Combination use: Adults: Gemcitabine in combination with cisplatin has been investigated using two dosage regimens. One regimen used a three week schedule and the other used a four week schedule.

The three week schedule used gemcitabine 1,250 mg/m^2, given by 30 minute intravenous infusion, on days 1 and 8 of each 21 day cycle. The three week schedule used cisplatin 75 - 100 mg/m^2 on day 1 of each 21 day cycle, administered before the gemcitabine dose. Dosage reduction with each cycle or dose omission within a cycle may be applied based upon the amount of toxicity experienced by the patient.

The four-week schedule used gemcitabine 1,000 mg/m^2, given by 30 minute intravenous infusion, on days 1, 8, and 15 of each 28 day cycle. The four week schedule used cisplatin 75 – 100 mg/m^2 on day 1 of each 28 day cycle, administered after the gemcitabine dose. Dosage reduction with each cycle or dose omission within a cycle may be applied based upon the amount of toxicity experienced by the patient.

Pancreatic cancer:
Adults: The recommended dose of gemcitabine is 1,000 mg/m^2, given by 30 minute intravenous infusion. This should be repeated once weekly for up to 7 weeks followed by a week of rest. Subsequent cycles should consist of injections once weekly for 3 consecutive weeks out of every 4 weeks. Dosage reduction with each cycle or dose omission within a cycle may be applied based upon the amount of toxicity experienced by the patient.

Bladder cancer:
In patients with bladder cancer who cannot tolerate cisplatin-based combinations, gemcitabine monotherapy should be considered a treatment option.

Single-agent use:
Adults: The recommended dose of gemcitabine is 1,250 mg/m^2, given by 30 minute intravenous infusion. The dose should be given on Days 1, 8 and 15 of each 28 day cycle. This four week cycle is then repeated. Dosage reduction with each cycle or dose omission within a cycle may be applied based upon the amount of toxicity experienced by the patient.

Combination use:
Adults: The recommended dose for gemcitabine is 1,000 mg/m^2, given by 30 minute intravenous infusion. The dose should be given on days 1, 8 and 15 of each 28 day cycle in combination with cisplatin. Cisplatin is given at a recommended dose of 70 mg/m^2 on Day 1 following gemcitabine or Day 2 of each 28 day cycle. This four week cycle is then repeated. Dosage reduction with each cycle or dose omission within a cycle may be applied based upon the amount of toxicity experienced by the patient. A clinical trial showed more myelosuppression when cisplatin was used in doses of 100 mg/m^2.

Breast cancer:
Adults: Gemcitabine in combination with paclitaxel is recommended using paclitaxel (175 mg/m^2) administered on Day 1 over approximately 3 hours as an intravenous infusion, followed by gemcitabine (1250 mg/m^2) as a 30-minute intravenous infusion on Days 1 and 8 of each 21 day cycle. Dose reduction with each cycle or within a cycle may be applied based upon the amount of toxicity experienced by the patient.

Ovarian cancer:
Adults: Gemcitabine in combination with carboplatin is recommended using gemcitabine 1000 mg/m^2 administered on Days 1 and 8 of each 21-day cycle as a 30-minute intravenous infusion. After gemcitabine, carboplatin should be given on Day 1 consistent with target AUC of 4.0 mg/mL/min. Dosage reduction with each cycle or within a cycle may be applied based upon the amount of toxicity experienced by the patient.

Dose Reduction:
Haematological Toxicity: Patients receiving gemcitabine should be monitored prior to each dose for platelet, leucocyte and granulocyte counts and, if there is evidence of toxicity, the dose of gemcitabine should be reduced or withheld.

Patients receiving gemcitabine should have an absolute granulocyte count of at least 1.5 ($x10^9$/L) and a platelet count of ≥ 100 ($x10^9$/L) prior to initiation of a cycle. Dose modifications of gemcitabine on Day 8 and/or Day 15 for haematological toxicity should be performed according to the guidelines below (Tables 2-4).

Gemcitabine monotherapy or in combination with cisplatin:

Table 2: Dose Modification of Gemcitabine on Day 8 and/or Day 15 for Gemcitabine Monotherapy or in Combination with Cisplatin

Absolute Granulocyte Count ($x10^9$/L)		Platelet Count ($x10^9$/L)	% of full dose
> 1.0	and	> 100	100
0.5 - 1.0	or	50 - 100	75
< 0.5	or	< 50	Hold*

* Treatment may be reinstated on Day 1 of the next cycle.

Gemcitabine in combination with paclitaxel:

Table 3: Dose Modification of Gemcitabine on Day 8 for Gemcitabine in Combination with Paclitaxel

Absolute Granulocyte Count ($x10^9$/L)		Platelet Count ($x10^9$/L)	% of Day 1 Gemcitabine Dose
≥ 1.2	and	> 75	100
1.0 - < 1.2	or	50 - 75	75
0.7 - < 1.0	and	≥ 50	50
< 0.7	or	< 50	Hold*

* Treatment may be reinstated on Day 1 of the next cycle.

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Gemcitabine in combination with carboplatin:

Table 4: Dose Modification of Gemcitabine on Day 8 for Gemcitabine in Combination with Carboplatin			
Absolute Granulocyte Count ($x10^9$/L)		**Platelet Count ($x10^9$/L)**	**% of Day 1 Gemcitabine Dose**
≥ 1.5	and	≥ 100	100
1.0 - < 1.5	or	75-99	50
< 1.0	or	< 75	Hold*
* Treatment may be reinstated on Day 1 of the next cycle.			

Other Toxicity: Periodic physical examination and checks of liver and kidney function should be made to detect non-haematological toxicity. Dosage reduction with each cycle or dose omission within a cycle may be applied based upon the amount of toxicity experienced by the patient. Doses should be withheld until toxicity has resolved in the opinion of the physician.

Gemcitabine is well tolerated during the infusion, with only a few cases of injection site reaction reported. There have been no reports of injection site necrosis. Gemcitabine can be easily administered on an outpatient basis.

Elderly Patients: Gemcitabine has been well tolerated in patients over the age of 65. There is no evidence to suggest that dose adjustments are necessary in the elderly, although gemcitabine clearance and half-life are affected by age.

Hepatic and Renal Impairment: Gemcitabine should be used with caution in patients with hepatic insufficiency or with impaired renal function. Dose reduction is recommended in patients with elevated serum bilirubin concentration because such patients are at increased risk of toxicity.

There are no safety or pharmacokinetic data available for this combination (gemcitabine and cisplatin) in patients with creatinine clearance < 60 mL/minute.

Children: No sufficient data to establish the efficacy and safety of gemcitabine in children.

OVERDOSE:

The symptoms of overdosage are likely to be an extension of the pharmacological actions of gemcitabine. Possible symptoms of toxicity are those listed under Adverse Effects. Haematopoietic, gastrointestinal, hepatic or renal toxicity may be seen depending on the dosage given and the physical condition of the patient. Toxicity may be delayed and life threatening (e.g. myelosuppression).

There is no antidote for overdosage of gemcitabine. Myelosuppression, paresthesias, and severe rash were the principal toxicities seen when a single dose as high as 5.7 g/m^2 have been administered by IV infusion over 30 minutes every two weeks with clinically acceptable toxicity. In the event of suspected overdose, the patient should be monitored with appropriate blood counts and should receive supportive therapy, as necessary.

HANDLING AND DISPOSAL:

Handling

The normal safety precautions for cytostatic agents must be observed when preparing and disposing of the infusion solution. Pregnant personnel should not handle the product. Handling of the Injection should be done in a safety box and protective coats and gloves should be used. If no safety box is available, the equipment should be supplemented with a mask and protective glasses.

If the preparation comes into contact with the eyes, this may cause serious irritation. The eyes should be rinsed immediately and thoroughly with water. If there is lasting irritation, a doctor should be consulted. If the solution is spilled on the skin, rinse thoroughly with water.

Instructions for dilution

The approved diluent for dilution of Gemcitabine Injection 38 mg/ml is sodium chloride 9 mg/ml (0.9%) solution for injection (without preservative).

Total quantity of Gemcitabine Injection required for an individual patient must be diluted, before use, to at least 500 ml of sodium chloride 9 mg/ml solution for injection to obtain clinically relevant concentrations. Further dilution with the same diluent can be done.

Based on the recommended dose (1000 mg/m^2 and 1250 mg/m^2) and body surface area (between 1.0 m^2 to 2.0 m^2) a concentration range of 2 mg/ml to 5 mg/ml is obtained.

The following instructions for dilution should be strictly followed in order to avoid adverse events.

1. Use aseptic technique during dilution of gemcitabine for intravenous infusion administration.
2. Parenteral medicinal products should be inspected visually for particulate matter and discolouration prior to administration. If particulate matter is observed, do not administer.

Any unused product or waste material should be disposed of in accordance with local requirements.

STORAGE:

Store below 30°C. Do not refrigerate or freeze.

Shelf life after dilution:

Chemical and physical in-use stability after dilution in 9 mg/ml sodium chloride solution at a concentration of 0.1 mg/ml and 5 mg/ml has been demonstrated for 7 days at 2°C to 8°C or at 25°C.

From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would normally not be longer than 24 hours at 2°C to 8°C, unless dilution has taken place in controlled and validated aseptic conditions.

PRESENTATION:
Gemcitabine Injection (GEMITA® RTU) 200 mg/5.26 ml: In a 6 ml USP type-I, tubular clear colorless glass vial.
Gemcitabine Injection (GEMITA® RTU): 1 g/26.3 ml: In a 30 ml USP type-I, tubular clear colorless glass vial.
Gemcitabine Injection (GEMITA® RTU): 2 g/52.6 ml: In a 100 ml USP type-I, molded clear colorless glass vial.

MANUFACTURES INFORMATION:

Manufactured in India by:
Fresenius Kabi Oncology Ltd.
Village Kishanpura
P.O. Guru Majra
Tehsil-Nalagarh
Distt. Solan (H.P) – 174101

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